

Interoperable and Modular System Architecture for Green-Energy Unmanned Surface Vehicles in Ocean Science

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Abstract—This article presents the open system architecture design of a modular electronic system used onboard unmanned surface vehicles. The developed solution enables this class of marine vehicles to accomplish marine research operations by ensuring safe long-endurance autonomous navigation as well as data gathering and transmission to shore. The onboard electronic system features modularity and scalability, making it easier to add navigational or scientific sensors, as well as to enhance software functionalities by adding new modules remotely. Our concept is currently field-tested on two wave-propelled and solar-powered surface vehicles equipped with multiple payloads, serving as in-situ sampling and data providers for oceanographers and biologists. The article describes the general system architecture and two different configurations in the two vessels that target phytoplankton and zooplankton, respectively.

Index Terms—Unmanned Surface Vehicle, System Architecture, Experimental Results

I. INTRODUCTION

Given the urgency of tracking fast-evolving environmental changes, innovations in autonomous robotic systems can greatly enhance our capacity for sustained and precise ocean monitoring [1]. By replacing costly, labor-intensive, and often hazardous human-led expeditions, unmanned surface vehicles (USVs) enable safer, more precise, and extended monitoring capabilities in challenging environments. These robotic platforms can operate continuously in adverse weather and remote areas, providing high-resolution data across spatial and temporal scales that would be impractical for human crews.

One application of autonomous USVs is the study of dynamic and often unpredictable phenomena such as algal blooms, which have significant ecological, economic, and health impacts worldwide. In Norway's fjords and coastal areas, harmful blooms can devastate marine life and affect local communities. Understanding these events—and other climate-affected processes like primary productivity, zooplankton and fish dynamics, as well as oxygen depletion, requires robust, real-time data. Many of these phenomena are driven or influenced by human activities, such as nutrient runoff or climate change, while others are natural processes fundamental to ecosystem health. Persistent robotic monitoring offers unparalleled insight into these events, helping researchers respond to

environmental threats and deepen our understanding of marine ecosystems.

This paper describes the design and implementation of a USV-based monitoring system architecture aimed at advancing our ability to study these phenomena. By integrating specialized sensors and a comprehensive data acquisition system, this autonomous platform addresses the need for high-precision, enduring, and real-time monitoring of coastal and fjord environments in Norway.

To date, oceanic exploration and monitoring of the upper water column, driven by scientific hypothesis and by means of long-endurance robotic platforms, has already been demonstrated, e.g. [2], [3]. Ocean exploration and monitoring have increasingly leveraged long-endurance, renewable-energy-powered surface vehicles, which have been developed primarily by private companies to meet various commercial and operational needs. Examples of such platforms include the Liquid Robotics WaveGlider, Offshore Sensing SailBuoy, AutoNaut, SailDrone, and the L3Harris C-Enduro. Each of these systems offers a unique architecture tailored to specific



Fig. 1. AutoNaut USV during operations in Nordfjord (Western Norway).

missions, such as extended environmental monitoring, security and defense operations, and oil and gas surveying. However, as proprietary products, they are not typically designed with the flexibility needed for academic research applications, where interoperability across a wide range of scientific equipment and tasks is essential.

In contrast, academic research platforms like the one presented in this article require open, modular, and scalable system architectures that prioritize customization, which is essential for supporting a diverse range of scientific projects. In an academic environment, it is critical that the platform can be easily adapted for different research objectives—often simultaneously or sequentially across projects. This adaptability allows us to reuse the same platform for various tasks, from ocean studies and marine biology to autonomy and control engineering, enhancing both flexibility and cost-effectiveness.

Furthermore, the open, modular and scalable design supports the active involvement of students, who contribute by developing specialized software and hardware modules. This collaborative approach reinforces the platform’s modularity and scalability as essential qualities, ensuring that each new contribution strengthens the system’s utility across multiple projects. Despite the increasing presence of commercially developed surface vehicles, there is a clear gap in the literature describing research-driven, long-endurance platforms that prioritize such interoperability, particularly for oceanographic research.

The AutoNaut USV [4] stands out as a green and versatile platform due to its unique wave-propelled design, which eliminates the need for fuel-based propulsion. The vehicle is provided with a propulsion system that relies on sea surface waves [5]. Two pairs of spring-loaded submerged hydrofoils are connected at the bow and stern by two vertical struts. When a surface wave lifts the bow or the stern of the vehicle, the corresponding strut lifts the foils, which are subsequently pulled back by the spring generating a forward thrust. This allows for extended deployment durations with minimal energy consumption, making it a powerful asset for marine science and environmental monitoring. Therefore, at the Norwegian University of Science and Technology (NTNU, Norway), the AutoNaut has been at the forefront of several interdisciplinary research projects focused on the Norwegian coastal and fjord regions. Leveraging its ability to operate autonomously for weeks without retrieval, we have employed the AutoNaut in diverse applications including fish migration studies in western Norway, providing real-time data on movement patterns and habitat usage [6]; monitoring of algal blooms in fjords, aiding in understanding their dynamics and impacts on local ecosystems [7]–[9]; integration with remote sensing data, bridging in-situ observations with hyperspectral satellite imagery for comprehensive environmental assessments [10], [11]. Moreover, the unique propulsion system of the AutoNaut USV opened the doors to innovative research in the field of autonomous systems, mission management, autopilot design and automatic collision avoidance [12]–[14]. Overall, the AutoNaut USV is serving as a general-purpose data collection platform, gathering a variety of physical, biological, and atmospheric parameters. The vehicle’s passive wave propulsion system and low-power electronics architecture have been instrumental in

achieving these outcomes, significantly reducing operational costs and enhancing safety compared to conventional methods. These achievements underline the AutoNaut’s utility in both fundamental research and applied oceanography, as described in [15].

Recognizing the need for enhanced capabilities, NTNU developed an open source system architecture with a custom onboard electronics system tailored to the AutoNaut’s unique requirements [16]. The first version of this system integrated autopilot functionalities for mission execution; a data collection system capable of interfacing with a wide array of sensors; a data transmission system for near real-time communication with shore stations. This architecture proved robust and reliable in all deployments. Building on this success, we designed and deployed a second-generation open-source onboard electronics system in 2023, focusing on lower complexity, compactness, higher performance, and modularity. The upgraded system integrates advanced components to support higher data acquisition rates, improved computational capabilities, and more efficient energy management.

We aim to present this second-generation architecture in detail, highlighting its design principles, enhancements, and compatibility with the AutoNaut platform. Recent field trials conducted in the Norwegian coasts and fjords have demonstrated its superior performance in complex marine environments. These trials involved simultaneous measurements of physical, chemical, and biological parameters, showcasing the system’s capacity for real-time, high-resolution data collection onboard two 5-meters long AutoNaut vehicles that are configured for two different science mission profiles: 1) phytoplankton observation and sampling, and 2) zooplankton and general ocean observations.

This article outlines the evolution of our onboard electronics system, emphasizing its role in advancing marine research. By sharing insights from our experiences, we hope to contribute to the broader adoption and development of autonomous technologies in oceanographic studies.

II. CONOPS AND SYSTEM REQUIREMENTS

This platform is designed to accommodate two primary operational profiles: 1) long-duration missions in open ocean and exposed coastal waters, and 2) short-duration missions within confined and sheltered coastal waters and fjords. These profiles cater to different research and testing needs, each presenting distinct challenges and necessitating tailored capabilities.

A. Exposed ocean and coastal missions

Long-duration deployments target the open ocean and coastal areas, where environmental conditions are more extreme. These missions are characterized by exposure to large waves, strong winds, and limited communication infrastructure, with satellite links being the primary means of data transfer. Due to communication constraints, onboard autonomy must be robust, allowing the vehicle to operate independently for extended periods of several hours or even days without parameter adjustments. The limited real-time data transfer capacity requires the system to preemptively plan and monitor energy and mission resources. The open ocean typically presents fewer obstacles, such as ship traffic, but demands a

durable and self-sufficient system that can withstand weeks or even months at sea.

B. Confined and sheltered coastal operations

Short-duration missions in fjords present their own set of challenges, including higher traffic density (e.g. coastal shipping and ferry routes), narrow waterways, and strong tidal currents. In addition, complex bathymetry and numerous obstacles, such as shallows, debris, islands, and buoys, require the vehicle to be highly reactive to sudden changes. In fjords, 4G connectivity may provide continuous high-bandwidth communication, enabling live streaming and remote control. This real-time link allows for rapid intervention, either remotely or through physical retrieval, as necessary. Such missions are typically limited to a few days. Fjord missions can also serve as testing grounds for functionalities before deploying the vehicle on extended ocean missions.

C. Requirements

Given these mission profiles, the following design requirements were defined:

- **Robustness:** The vehicle must be engineered to operate for months with minimal maintenance. Mechanically, this means a durable, watertight, and streamlined design with a limited number of moving parts. Electronically, all components, cables, connectors, and sensors, should be of industrial quality to ensure resilience. In addition, the control system should provide reliable fallback options, with redundant navigation sensors to guarantee functionality even under adverse conditions. Although the proposed control system architecture is simplified compared to our previous version, most elements of previous risk analyses [9], [17] are still relevant and used as a basis for the new design.
- **Energy management:** A reliable energy management system is essential, providing the capability to plan, monitor, and optimize energy consumption. The system must support solar energy harvesting for extended missions. A low-level control system enables scheduling of power to specific components, allowing selective activation and deactivation to maximize efficiency over prolonged deployments.
- **Communication:** The vehicle requires three distinct types of communication links. A near-real-time, low-bandwidth link with global coverage serves to continuously monitor health and location data, as well as offer basic emergency control. A real-time, intermediate-bandwidth link with regional coverage enables more detailed vehicle telemetry and control, as well as software upgrades while in service. Lastly, a high-bandwidth, short-range link allows for real-time sensor data transmission. Mission scenarios will dictate which links are active, with each faster link able to cover the functionality of slower links to ensure uninterrupted data flow across different ranges and environments.
- **Autonomy and Control:** The system must support autonomous navigation and decision-making, calculating trajectory adjustments based on environmental factors such as sea currents and winds. The platform should

accommodate advancements in control algorithms, autonomy, and AI, and be compatible with multi-vehicle, cooperative mission frameworks for future scalability. In addition to autonomous operations, the vehicle should support manual control for precise operations during launch and recovery.

- **Modularity and Scalability:** Both the control and payload systems should be designed for modularity, allowing each subsystem to be upgraded independently. This flexibility supports continuous advancements in each area without imposing limitations on others. Additionally, the platform should be designed to accommodate new scientific instruments in the future, allowing researchers to expand its capabilities to meet emerging scientific requirements.

III. ARCHITECTURE OVERVIEW

A. Propulsion and steering

The self-propelling mechanism of the AutoNaut USV limits the maximum speed achieved during operations to approximately 3 knots, depending on the sea-state, ocean current, and the configuration of the propulsion system (see Figure 2). The bow strut employs a piston propulsion mechanism that facilitates synchronous torsion of the hydrofoils, while the aft strut features independent spring-loaded hydrofoils. This design is effective in different sea conditions: while short, fast fjord waves induce significant rolling of the vehicle, independent springs are capable of generating greater thrust compared to a single piston under the same rolling conditions. This advantage allows the USV to maintain higher speeds when navigating fjords.

However, the independent spring mechanism has limitations; it is inherently weaker and less robust than the piston system. In open ocean environments characterized by larger and slower waves, the piston propulsion mechanism excels, delivering higher thrust and speed while being more resilient to stress. The robust design of the piston system ensures reliable performance in challenging conditions, where consistent thrust is crucial for extended operations.

The heading of the vehicle is controlled using a rudder commanded by the navigation control unit, which allows for turns of up to $\pm 45^\circ$ relative to its centered position. Additionally, the platform is equipped with a small thruster, which can be activated manually or by the collision avoidance algorithm. This feature enables sharper maneuvers and provides additional propulsion whenever surface waves are insufficient to generate acceptable thrust.

B. Electric energy and power management system

The AutoNaut USV is designed with a robust energy harvesting and management system. Three 110 Wp solar panels provide the primary source of renewable energy, charging a bank of four 105 Ah, 12V Lithium batteries arranged in parallel. These batteries ensure a stable and sufficient power supply to meet the system's needs throughout extended deployments. A Maximum Power Point Tracking (MPPT) regulator is employed to optimize the charging process, ensuring that the solar panels operate at peak efficiency under varying light conditions. The MPPT regulator not only charges the battery bank but also supplies a 12V output that powers the onboard

electronic system, as illustrated in Figure 3. The MPPT is configured with a protective feature: it disconnects the load when the battery voltage falls below a user-defined threshold to prevent deep discharge and battery damage, and it reconnects the load once the batteries have been sufficiently recharged. This safeguard enhances the longevity and reliability of the battery system.

The electronic system onboard, which handles navigation, control, data acquisition and transmission, is equipped with power management capabilities. It can selectively power down non-essential electrical components when energy reserves are low, thereby optimizing power consumption according to operational demands. This dynamic power management is critical during low-energy conditions or when solar charging is insufficient, ensuring that the most vital systems remain operational. In idle conditions - when the thruster is inactive and the rudder is not compensating for significant forces - the USV's average current consumption is approximately 0.7A. However, energy consumption can vary significantly depending on mission activities. For example, bringing the thruster to full speed can increase the current consumption to as much as 15A. The energy cost associated with thruster use is particularly pronounced during maneuvers requiring rapid acceleration or when countering strong environmental forces. Similarly, the power consumption of the rudder is closely related to its operational profile. Frequent and rapid adjustments to correct for small deviations in heading or course - for example in high sea conditions - result in higher energy usage. By implementing a filtered control strategy that reduces unnecessary corrections in the absence of strong currents, energy consumption can be minimized. This approach not only improves efficiency, but also prolongs mission duration by conserving battery power, as discussed in [12].

Energy autonomy calculations: In scenarios where sunlight is absent, the USV must rely solely on its battery reserves. Given the total battery capacity of 5040 Wh, the operational autonomy can be estimated for various power consumption scenarios:

- **Idle conditions:** With an average consumption of 8.4 W:

$$\text{Autonomy} = \frac{5040 \text{ Wh}}{8.4 \text{ W}} \approx 600 \text{ hours},$$

which corresponds to roughly 25 days of continuous operation.

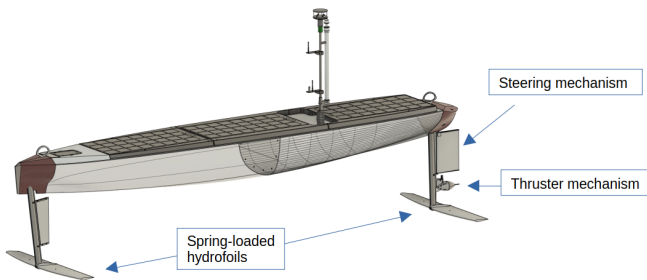


Fig. 2. An overview of the propulsion and steering mechanisms.

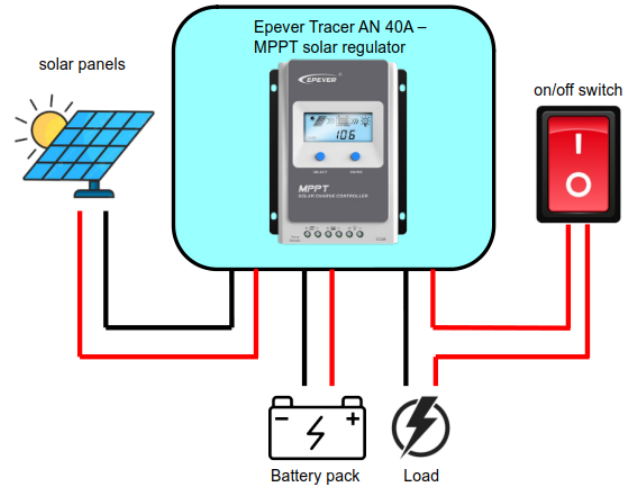


Fig. 3. Energy-harvesting system overview.

- **Moderate operation:** Assuming an average consumption of about 60 W (approximately 5 A at 12 V):

$$\text{Autonomy} = \frac{5040 \text{ Wh}}{60 \text{ W}} \approx 84 \text{ hours},$$

or roughly 3.5 days.

- **High-demand scenarios:** Under worst-case conditions, with a current consumption of 15 A (approximately 180 W):

$$\text{Autonomy} = \frac{5040 \text{ Wh}}{180 \text{ W}} \approx 28 \text{ hours}.$$

These calculations underscore the importance of solar charging for extending mission durations. Even modest sunlight can significantly supplement the energy budget, enabling the USV to operate well beyond the limits imposed by battery capacity alone.

C. Embedded electronic system

The onboard electronic system of the AutoNaut USV has been designed from off-the-shelf (COTS) components to achieve the critical functionalities of autonomous navigation, data collection, storage, and transmission over various links (Wi-Fi, LTE, and satellite). Given the harsh marine environment, the system is engineered for robustness and reliability. To meet these requirements, we have selected a suite of Embedded TS boards from Technologic Systems, which are known for their rugged design and low power consumption. At the core of our system is the TS-7250-V3 single-board computer (see Figure 4). This board is particularly attractive because of its exceptionally low power consumption - typically operating at less than 1.5 watts under normal conditions - making it ideal for long-endurance missions where energy efficiency is paramount.

To expand the capabilities of the TS-7250-V3 and meet the diverse communication requirements of onboard sensors and actuators, we integrated several peripheral boards from Technologic Systems, all of which interface with the main SBC via the PC/104 expansion bus as depicted in Figure 4:

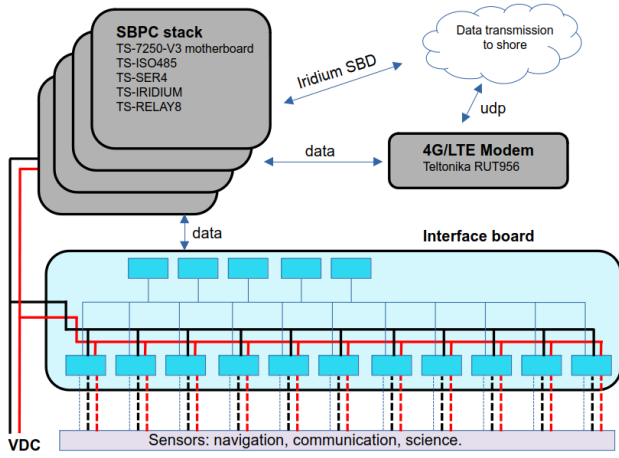


Fig. 4. Simplified representation of the onboard electronic system. A stack of boards features the CPU, RAM and various interfaces for serial communication; these are interconnected to an interface board that separates all communication lines for each sensor and adds power.

- **TS-ISO485:** This board provides isolated RS-485 serial communication, ensuring reliable and noise-resistant data exchange with sensors and actuators that use this protocol. Electrical isolation is particularly important in the marine environment to protect against ground loops and electromagnetic interference.
- **TS-SER4:** A four-port RS-232/RS-422/RS-485 serial expansion board, allowing seamless connection to multiple sensors and devices that rely on standard serial communication. This expansion increases the number of available serial ports beyond what the main SBC provides.
- **TS-IRIDIUM:** A dedicated Iridium 9602 satellite module that enables global data transmission through the Iridium satellite network. This is crucial for mission scenarios where LTE coverage is unavailable, ensuring continuous monitoring and remote control capabilities.
- **TS-RELAY8:** An eight-channel relay output board that allows the system to control the power to various onboard components, such as sensors, communication devices, and auxiliary systems. This capability is essential for energy management, as it enables the selective activation and deactivation of power-hungry devices depending on the mission requirements.

To efficiently manage the connections between the TS-7250-V3 stack and the various onboard sensors, we developed a custom breakout board (bottom part of Figure 4). This board serves a dual purpose:

- **Signal Routing:** It organizes and distributes the communication lines from the full TS stack, ensuring that each sensor receives the appropriate data interface.
- **Power Distribution:** It provides regulated power to sensors and peripherals, with selectable voltage levels (3.3V, 5V, and 12V) configured via jumpers. Each sensor is connected via a dedicated port that includes both power and communication lines, simplifying wiring and ensuring modularity.

This modular architecture was specifically designed to meet

the diverse protocol requirements of the onboard sensors while maintaining robustness, ease of maintenance, and power efficiency.

D. Guidance, navigation and control (GNC) system and collision avoidance

The AutoNaut USV is equipped with a robust onboard GNC system that supports both autonomous navigation and collision avoidance. Central to this system is the TS-7250-V3, running Linux Debian 12. The various software packages running on the TS-7250-V3 handle continuous communication with shore stations and execute the GNC algorithms necessary for autonomous operation.

The primary onboard GNC software is DUNE, developed by LSTS at UPorto, [18]. DUNE manages the vehicle's course by controlling the rudder (and thruster) while also gathering, storing, and transmitting data from various scientific and navigational sensors. In combination with the Neptus command and control interface, used by operators to observe and control the USV, the DUNE architecture facilitates real-time monitoring and control, which are essential in the dynamic and often unpredictable marine environment. Heading or course-over-ground (COG) control is achieved through an advanced autopilot system, illustrated in Figure 5. This system receives heading and COG measurements from a Hemisphere GPS compass and computes the appropriate rudder angle to minimize the deviation between the measured and intended heading or course. The current autopilot builds upon previous research; for example, earlier work detailed in [12] presented a course control system based on a comprehensive model of the USV's dynamics, and subsequent improvements introduced gain-scheduling functionalities - adjustments based on real-time measurements - to enhance performance under varying conditions as described in [13]. These developments allow the autopilot to dynamically adapt to changing environmental factors, ensuring more accurate and stable navigation.

In addition to its navigational capabilities, the AutoNaut USV incorporates a collision avoidance module (see Figure 5) as described in [14], [19]. This module (named AC/AG system in the figure) processes data from an onboard AIS transceiver and from digital charts stored onboard to detect obstacles and adjust the vehicle's trajectory accordingly. The collision avoidance system is designed to quickly respond to sudden changes in the vehicle's surroundings, making it particularly effective in confined coastal areas or regions with high vessel traffic. In emergency situations, the collision avoidance algorithms can override the pre-planned navigation path, allowing the USV to execute evasive maneuvers. Furthermore, the system supports a seamless transition to manual control when necessary, ensuring higher levels of operational safety.

E. Communication to shore

The USVs are equipped with a comprehensive communication suite designed to ensure continuous connectivity under diverse conditions. The primary link is 4G/LTE, widely available in Norwegian fjords and coastal waters, which allows for high-frequency data transmission and a constant live feed from the USV. When LTE coverage is lost, the onboard software automatically detects the disconnection and switches

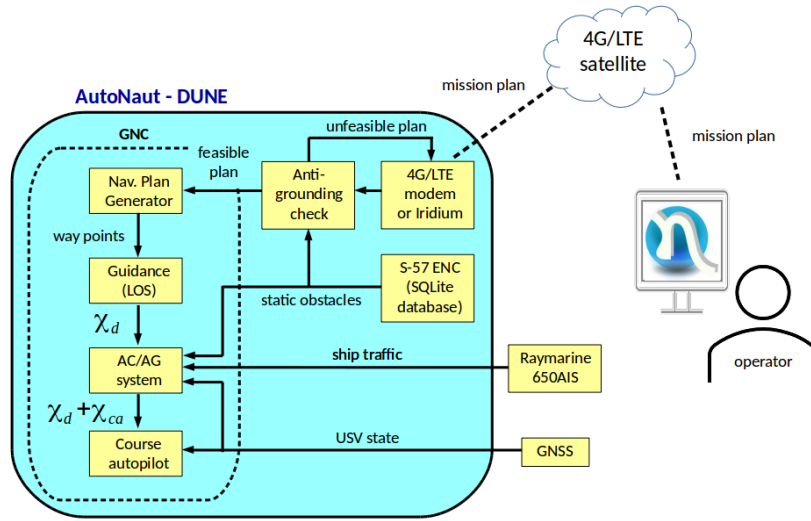


Fig. 5. GNC with collision avoidance architecture overview [19].

to satellite communication. Additionally, a local WiFi radio provides high-speed data transfers when the operator is in close proximity.

1) *4G/LTE link*: The LTE modem chosen for this application is the Teltonika RUT956, selected for its exceptional configuration flexibility. This modem makes use of a SIM card and features automatic Dynamic DNS (DDNS) update, ensuring that the USV is always reachable via a chosen domain name. Its firewall is configured to allow bidirectional communication exclusively with NTNU servers, meaning operators must connect through the NTNU network to control the vehicle or fetch data. Beyond standard 4G/LTE connectivity, the RUT956 offers additional features such as the capability to remotely control and perform hard reboots on hardware components via its input/outputs. Additionally, it routes network traffic to local devices like the onboard IP camera and the WiFi radio. With NAT and port forwarding rules in place, the modem secures the USV's local network against external attacks. Furthermore, the RUT956 supports ZeroTier - a software-defined networking solution that facilitates secure communication channels between two or more networks.

2) *Satellite link*: For areas outside LTE coverage, the USV employs an Iridium 9602 module that uses the Short Burst Data (SBD) protocol. The SBD protocol is designed to transmit small data packets - where one credit corresponds to 50 bytes - and is widely used in remote monitoring applications. While it offers proven global coverage, the Iridium SBD service is relatively expensive, with costs incurred on a per-credit basis (and a monthly service fee). To limit these costs, the system automatically sends only essential mission reports and critical parameters such as battery voltage, the average error between the desired and measured course over ground, the USV speed over ground, and location. Operators can request more detailed data on-demand. It is worth noting that while SBD is robust and reliable, alternative satellite communication protocols may offer more economically attractive options with higher throughput, although they require more power consuming hardware setups.

3) *WiFi radio link*: The USV is equipped with a Ubiquiti Bullet AC radio (rated IP67 for outdoor use), which provides a high-speed WiFi connection when the operator is nearby. This radio is not active by default; it is remotely activated via commands sent over either LTE or satellite. Once turned on, the WiFi radio establishes an access point to the local network of the USV, that is ideal for transferring large amounts of data — such as detailed logs or video files — more quickly than over satellite or LTE. This functionality is also useful for on-site debugging, rapid command transmission, or interfacing with nearby systems on ships or moored buoys.

IV. VEHICLE CONFIGURATIONS

Both USV configurations share a standardized scientific payload that includes the SBE49 FastCat CTD, EcoPuck Triplet-w, Aandera Optode 4835, and Airmar 220WX weather station. These instruments were carefully selected because their combined data output supports a wide range of scientific studies by providing high-resolution measurements of key physical, chemical, biological, and meteorological parameters in marine environments.

- SBE49 FastCat CTD characterizes the water column's physical properties: conductivity (S/m), water temperature (°C), depth/pressure (hPa), salinity (PSU), sound speed (m/s).
- EcoPuck Triplet-w captures essential optical water indicators: chlorophyll fluorescence (µg/L), FDOM fluorescence (PPB), turbidity (NTU).
- Aandera Optode 4835 provides accurate measurements of dissolved oxygen levels: dissolved Oxygen concentration (µM), oxygen saturation (%), water temperature (°C).
- Airmar 220WX seather station collects critical meteorological data, such as wind speed (m/s), wind direction (°), air temperature (°C) and atmospheric pressure (hPa).

The two USV configurations employ specific sensors that target the study of phytoplankton and zooplankton, as described in the next two sections.

A. Phytoplankton monitoring and sampling

One USV is dedicated to phytoplankton observation and sampling and is equipped with a specialized payload that includes an Eco PAR radiometers¹, a McLane phytoplankton water sampler² (PPS), and two Trios Ramses radiometers³ - one oriented upward and the other downward. The Eco PAR sensor quantifies Photosynthetically Active Radiation in the upper water column, a key driver of phytoplankton photosynthesis, while the McLane sampler collects water samples for subsequent laboratory analysis of phytoplankton. Meanwhile, Trios Ramses radiometers provide complementary radiometric measurements from both above and below the water surface, providing insights into solar illumination, light attenuation, and spectral reflectance properties that affect phytoplankton distribution and productivity.

1) *Radiance and irradiance sensors:* The two Trios Ramses sensors are mounted to measure down-welling irradiance from the sun, and up-welling radiance from the water. As illustrated in Figure 6, a cone is attached to the instrument that measures upwelling radiance to block direct radiation from the sky [20]. The sensors are originally designed to be interfaced via RS232 from a proprietary program on a computer running Windows. Instead, the sensors are interfaced from the TS-7250-V3 computer running Linux, using a custom script that commands the sensor to perform a measurement and subsequently stores the data returned from the sensor. The data is then processed by the PyTriOs open-source Python program [21], producing calibrated water-leaving reflectance spectra.

2) *PPS integration:* The PPS features a patented multiport valve system that isolates individual samples and distributes a volume of water directly to a filter holder, passing through a pump before exiting the system. The instrument is equipped with 24 of these filter holders, which allow a total of 24 samples (events) to be collected, before the instrument needs servicing and sample retrieval. The instrument can feature antifouling and fixative reservoirs to inject solutions for cleaning and for sample preservation respectively during longer deployments. The version described here features both the antifouling and fixative additions. The fixative unit includes an extra valve that regulates the intake of water and the switch to a fixative solution channeled into the PPS tubing via the fixative bag. The fixative unit has an exhaust bag where the remaining fixative is stored after it passes and fills the filter holder containing the sample. The antifouling is injected via one of the 24 ports of the multiport valve, limiting the total number of samples to 23.

Due to the limited space inside the USV, the PPS underwent significant modifications to be physically integrated into the AutoNaut hull. The original version of the PPS features four units: the main unit (multiport valve, fixative valve, pump and filter plate with filter holders), a fixative unit (fixative bag and exhaust bag), the antifouling unit and the controller housing (control and power unit). All these units were regrouped



Fig. 6. Spectrometer with sky-blocking cone for measuring upwell radiance from the ocean, mounted on the side of the AutoNaut's hull.

and retrofitted to overcome the main space constraints. The following modifications were made:

- **Main unit:** the main challenge was to overcome the dimensions of the hatch that did not allow the insertion of the multiport valve together with the filter plate. The latter was redesigned and produced with reduced margins and separation gaps between the filter holders.
- **Fixative and antifouling units:** the antifouling unit was glued into the fixative unit container, making these into one unit and placed horizontally behind the main unit.
- **Controller housing:** the unit was reduced to one third of the original size and to a fraction of the weight by removing the batteries together with the pressure housing; the resulting plastic housing contains only the electronics and original connections to each of the PPS components.
- **Inlet and outlet:** the PPS water inlet and outlet had to be extended with tubing through two holes in the keel, located just beneath the main valve. These extensions are only about 40 cm long, in which the inlet was located at the bottom of the keel and projected forward (approx. 15 cm), while the outlet was located higher and pointing backward, to avoid sampling of the same exiting water.

The original PPS instrument is pre-programmed from a laptop, using the menu-driven software interface provided by the manufacturer (McLane Term). Sampling parameters are then pre-programmed, as well as the sampling frequency that allows a maximum deployment time. In this mode, the sampling programming is predetermined before deployment, with no possibility of modifying it.

¹<https://www.seabird.com/eco-photosynthetically-active-radiation-par-sensor/product?id=60762467733>

²<https://mclanelabs.com/phytoplankton-sampler/>

³<https://trios.de/en/ramses-g2/>

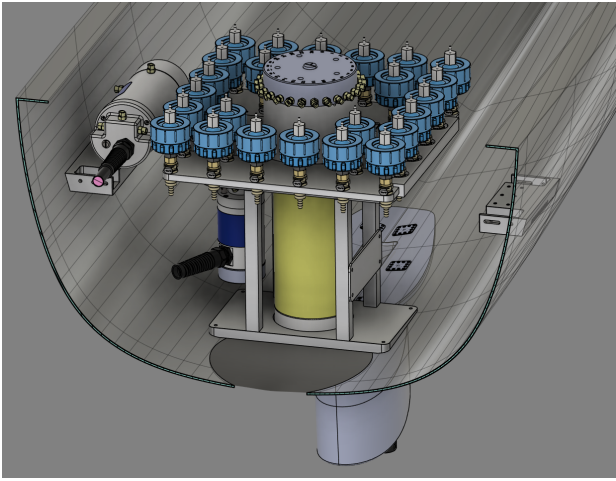


Fig. 7. An zoomed-out visual of the modified PPS in the USV's hull.

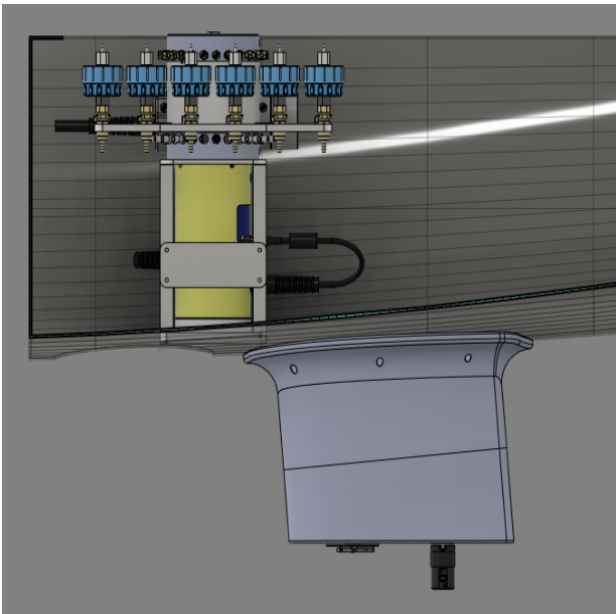


Fig. 8. Side view of the modified PPS inside the rear hull compartment.

Onboard the AutoNaut, the PPS is powered via a relay from the USV batteries. The PPS features a standard RS232 serial communication interface that allows the onboard system to send commands and receives information from the embedded PPS electronics. The AutoNaut USV control unit can connect to the serial line and send commands to start or stop a sampling event. This allows us to use the PPS for adaptive sampling applications, in which a sampling event is based on the variation of other measurements onboard the USV (e.g., chlorophyll concentration).

B. Zooplankton monitoring

The second USV is configured for zooplankton monitoring and is equipped with a Kongsberg EK80 transceiver and the Wideband Transceiver (WBT) Mini. The Kongsberg EK80 is a high-resolution acoustic instrument that provides de-

tailed echosounder data essential to profile the distribution and biomass of zooplankton, while the WBT Mini enhances detection sensitivity by capturing wide-band acoustic measurements. Together, these devices generate comprehensive datasets that allow researchers to study the dynamics of zooplankton, including vertical diel migration patterns and responses to environmental changes, thus deepening our understanding of the functioning of marine ecosystems. The experience with EK80 is that it provides data with low levels of noise, likely due to the low noise characteristics of the AutoNaut platform with passive foils for propulsion and with normal operations running without any rotating machinery.

The EK80 is operated through proprietary software from the vendor that runs on an embedded Windows computer. The computer is accessible through the modem and can be operated remotely.

V. DATA MANAGEMENT AND DISPLAY

Once the data are sent from the USVs, they are received by a centralized server at NTNU and made available to operators and involved scientists. Both 4G/LTE and Iridium data streams are injected into an InfluxDB (v1.8) database. The server firewall is configured to allow the AutoNaut USVs to transmit data directly to the remote database. To transmit data from AutoNaut USVs to the central database, the system onboard the USV uses cURL, a command-line tool that enables data transfer over various network protocols, including HTTP. InfluxDB v1.8 provides an HTTP API that allows direct data injection using cURL commands. Each data point is formatted according to InfluxDB's line protocol, which consists of a measurement name, optional tags, field values, and a timestamp. Figure 10 shows the transmission of the data via LTE/4G network and the reception from the remote server at NTNU. Once data are received, they are both served to Neptus and to a bash script that injects them into the InfluxDB database. While the transmission to Neptus is

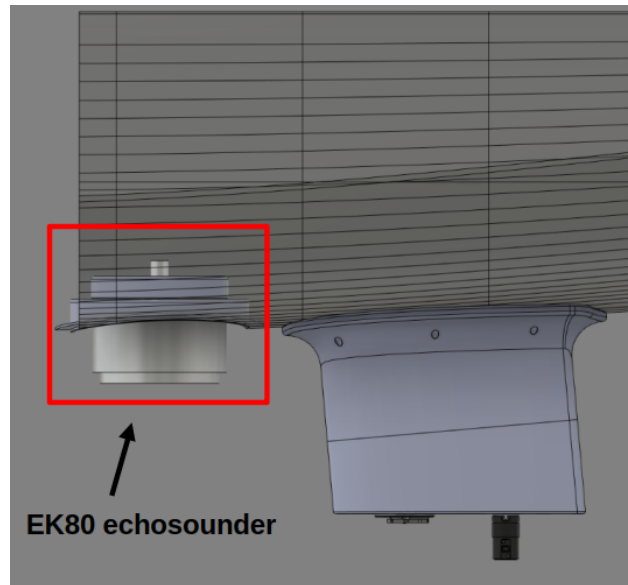


Fig. 9. Side view of the EK80 inside the hull of the USV.

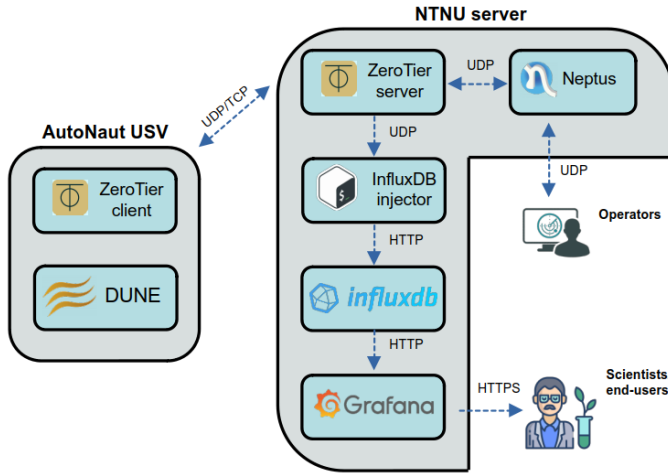


Fig. 10. 4G/LTE data pipeline from the USV to operators and scientists.

used to the operators to have a communication link to the USV, InfluxDB serves data to Grafana, which continuously queries the database and fills graphic dashboards with the latest received data.

A. Mission monitoring and control

The data received by the ZeroTier application running on the server are served instantly to Neptus via UDP network (see Figure 11). This allows the operator currently connected to the Neptus instance on the server to have a live feed of the mission. The operator can observe the mission status and all the USV parameters (potentially a lot more than those that are retained in the database). Moreover, the operator can update the mission profile, change the intended navigation plan and manually control components and sensors. On the server, two complementary services run continuously to make the collected data accessible and understandable.

B. Data display

Grafana (see Figure 12) is an open-source platform for data visualization and monitoring. It is particularly useful for

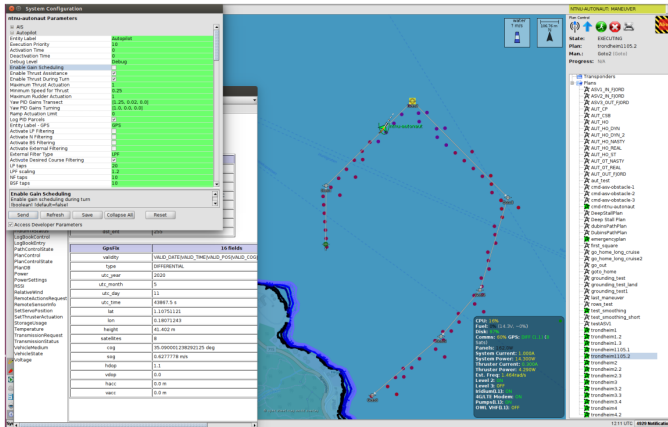


Fig. 11. Neptus command and control interface: USV during sea trials in Trondheimsfjord.

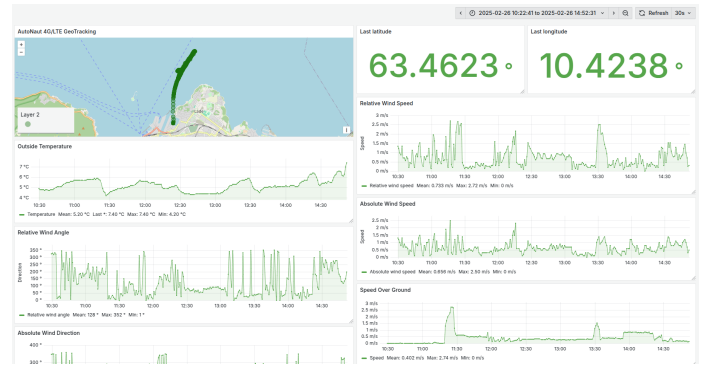


Fig. 12. Data display via Grafana during a field trial on February 26th 2025 in Trondheimsfjord.

this application because it periodically queries the InfluxDB database to retrieve up-to-date measurements. NTNU has developed several custom dashboards for each USV. These dashboards present real-time and historical data in an intuitive, graphical format, allowing operators and scientists to monitor mission status, analyze trends, and diagnose potential problems quickly. Access to these dashboards is secured and limited to authorized personnel, ensuring that sensitive operational data remain confidential.

C. Data download

In parallel with real-time display, a dedicated Python service is running on the server to facilitate data download and report generation, as depicted in Figure 13. This service uses a combination of JavaScript, HTML, and CSS to serve a user-friendly webpage on a specific port. Through this interface, end-users can:

- Select one of the two USVs.
- Choose specific sensor data of interest.
- Define the desired time range for the data.

Once the user makes these selections and clicks the corresponding button, the back-end code is executed to fetch the requested data from the InfluxDB database. The service then automatically builds a PDF report that includes graphs and statistics summarizing the selected data over the chosen time period. In addition, users have the option to download a CSV file containing the raw data, allowing further offline analysis or integration with other tools.

VI. CONCLUSIONS & FUTURE WORK

The article describes the general system architecture and two different configurations in the two vessels that target phytoplankton and zooplankton, respectively. Modularity, interoperability, system integration aspects, and robustness are highlighted.

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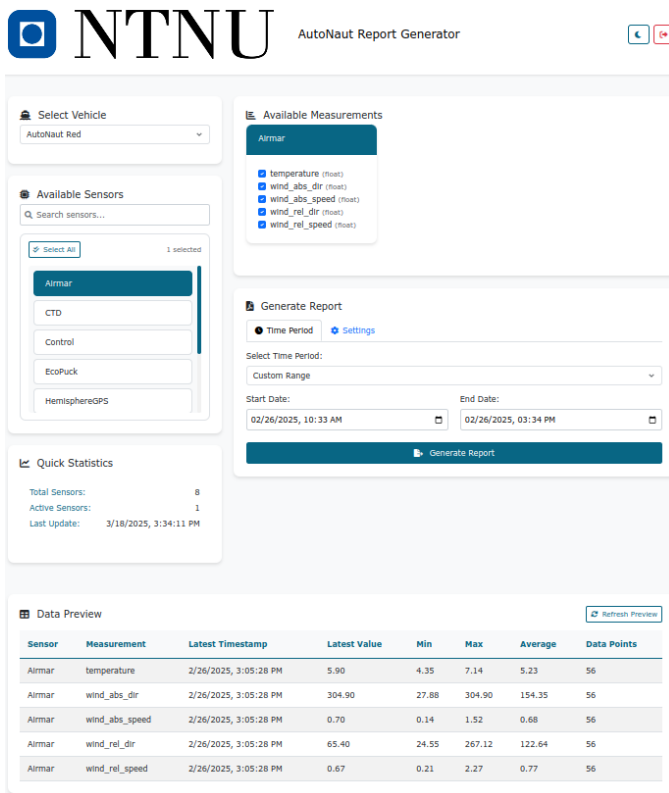


Fig. 13. Webpage for the generation of data reports and CSV files.

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